A DIFFERENTIAL RECURRENCE FOR THE ALTERNATING BINOMIAL TRANSFORM OF THE SECOND-ORDER EULERIAN POLYNOMIALS (OEIS A395454)

MATH0

Abstract. Let $P_m(t)=\sum_{k=0}^{m}T(m,k)t^k$ be the row polynomials of the triangle OEIS A395454, $T(n,k)=\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}E_2(j,n-k)$, where $E_2(j,i)$ are the second-order Eulerian numbers. We prove that $P_m=(m-1)(1+t)P_{m-1}+(t-t^2)P'_{m-1}+2(m-1)t\,P_{m-2}$ in $\mathbb{Z}[t]$ for every $m\ge 2$. This is the recurrence conjectured by Per W. Alexandersson in the OEIS entry (June 2026) when the rows are counted from 1; with the entry's own numbering from 0 the formula as written fails already at row 2 (and, as we show on paper, for every $n\ge 2$). We found no earlier proof, so we classify the result as new, although the argument is a short computation with reversed second-order Eulerian polynomials and their binomial transform. The result is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against all 55 listed terms.

1. Introduction

The second-order Eulerian numbers $E_2(n,k)$, in the indexing of Graham, Knuth and Patashnik [4, §6.2] and of the OEIS entry A201637 [10], are given by $E_2(0,k)=[k=0]$ and, for $n\ge 1$ and $k\ge 0$,

$$E_2(n,k)=(2n-1-k)\,E_2(n-1,k-1)+(k+1)\,E_2(n-1,k),\qquad E_2(n-1,-1)=0. \tag{1}$$

The first rows are (1), (1, 0), (1, 2, 0), (1, 8, 6, 0). In April 2026 Peter Luschny added to the OEIS the triangle A395454 [11],

$$T(n,k)=\sum_{j=0}^{n}\binom{-j-1}{-n-1}E_2(j,n-k)\qquad(0\le k\le n),$$

where the binomial coefficient at negative integers is understood as in Maple; for $0\le j\le n$ it equals $(-1)^{n-j}\binom{n}{j}$, so

$$T(n,k)=\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}E_2(j,n-k). \tag{2}$$

Thus the reversed rows of A395454 are the alternating binomial transform of the rows of A201637. The rows of A395454 begin (1), (0, 0), (0, 2, 0), (0, 6, 2, 0), (0, 24, 34, 2, 0); their sums $1,0,2,8,60,544,\ldots$ form A053871 [9], defined by $a(n)=2(n-1)\big(a(n-1)+a(n-2)\big)$, $a(0)=1$, $a(1)=0$. On June 27, 2026 Per W. Alexandersson added to the entry a conjectured three-term differential recurrence for the row polynomials (Section 2). The current revision (#30, October 14, 2026) still labels it a conjecture and records no proof. With the entry's numbering of the rows from 0 the formula is false, already for row 2; with the rows counted from 1, which is evidently what the author's $P(n)$ refers to, it holds. This is our main result (Theorem 1).

Date: October 24, 2026.

Prior work. The math0 referees found no earlier proof or refutation (searching OEIS cross-references, arXiv and the web). The closest work we know of is by Ma and Yeh [7] on Stirling permutations of the second kind and Stirling derangements, enumerated by cycle ascent plateaus and cycles. We use the numbering of the arXiv version; its Proposition 9, Theorem 17 and Corollary 18 are Proposition 4, Theorem 11 and Corollary 12 of the journal version. Their Theorem 17 is a recurrence of the same shape for the Stirling derangement polynomials, which we denote $D_n(x,q)$ (their $R_n(x,q)$) to avoid a clash with our R_j in Section 3: $D_{n+1} = 2nxD_n + 2x(1-x)\,\partial_x D_n + 2nxq\,D_{n-1}$ for $n \geq 1$. It concerns a different family (at $q=1$, $D_3 = 4x + 4x^2$, whereas $P_3 = 6t+2t^2$); compare Remark 3. Their Corollary 18 shows that the numbers of Stirling derangements satisfy the recurrence and initial values of A053871, so they are the row sums of A395454 (see Corollary 2). A referee observed that their Proposition 9 and binomial inversion over fixed points suggest that $P_n(t)$ is the distribution of $n -$ casc over Stirling derangements of order n (checked for $n = 2, 3$); we have not verified this further and do not use it. The related papers [1, 6] do not treat this triangle or recurrence, nor does Fried's recent collection of proofs of OEIS conjectures [3]. Our proof is routine: a differential recurrence for the reversed second-order Eulerian polynomials (a reformulation of (1)) is transported through the binomial transform (2); the result is new only in that we found no written proof.

2. Statement

The FORMULA section of A395454 [11] contains, verbatim, the line

```
Conjecture:  Let P(n) be the generating polynomial (in t) of row n.
Then P(n) = (n - 2*t + n*t - 2)*P(n-1) + (t - t^2)*P'(n-1) +
t*(2*n-4)*P(n-2).  - _Per W. Alexandersson_, Jun 27 2026
```

Here P' is the derivative in t, and $n - 2t + nt - 2 = (n-2)(1+t)$. For $m \geq 0$ let $P_m(t) = \sum_{k=0}^{m} T(m,k)t^k \in \mathbb{Z}[t]$ be the generating polynomial of the row with index m in the entry's numbering (offset 0), so $P_0 = 1$, $P_1 = 0$, $P_2 = 2t$, $P_3 = 6t + 2t^2$, $P_4 = 24t + 34t^2 + 2t^3$.

The reading of the conjecture. If $P(n)$ is taken to be P_n, the formula is false: at $n = 2$ its right-hand side is $0 \cdot P_1 + (t - t^2)P_1' + 0 \cdot P_0 = 0$, while $P_2 = 2t$; at $n = 3$ it gives $4t$ instead of $6t + 2t^2$ (see also Corollary 2). If $P(n)$ is the polynomial $\widetilde{P}(n) = P_{n-1}$ of the n-th row counted from 1, the formula holds. Exact computations by the math0 fidelity reviewer found the literal reading false for $2 \leq n \leq 19$ and the shifted one true for $3 \leq n \leq 20$; the row sums confirm the shift, since at $t = 1$ the conjecture is the recurrence of A053871 moved by one index. The math0 statement therefore formalizes the shifted reading for $n \geq 3$, where $\widetilde{P}(n), \widetilde{P}(n-1), \widetilde{P}(n-2)$ are all rows (at $n = 2$ it is $0 = 0$); the literal reading is recorded separately and refuted.

Theorem 1. *For every integer $m \geq 2$, in $\mathbb{Z}[t]$,*

$$P_m = (m-1)(1+t)\,P_{m-1} + (t - t^2)\,P'_{m-1} + 2(m-1)t\,P_{m-2}. \tag{3}$$

Equivalently (put $m = n-1$), with $\widetilde{P}(n) = P_{n-1}$, for every $n \geq 3$,

$$\widetilde{P}(n) = (n - 2t + nt - 2)\,\widetilde{P}(n-1) + (t - t^2)\,\widetilde{P}'(n-1) + t(2n-4)\,\widetilde{P}(n-2).$$

Corollary 2 (not formalized)**.** *Let $a(m) = P_m(1)$. Then $a(m) = 2(m-1)\big(a(m-1) + a(m-2)\big)$ for $m \geq 2$, so a is A053871. Moreover, for every $n \geq 2$ the literal reading $P_n = (n-2)(1+t)P_{n-1} + (t - t^2)P'_{n-1} + 2(n-2)t\,P_{n-2}$ is false.*

Proof. Put $t = 1$ in (3); with $a(0) = 1$, $a(1) = 0$ this is A053871. Induction gives $a(m) \geq 0$, so $a(2) = 2$ and $a(m) \geq 2(m-1)a(m-1) > 0$ for $m \geq 3$; thus $a(k) > 0$ for $k \neq 1$. Subtracting the literal right-hand side from (3) at $m = n$ leaves $(1+t)P_{n-1} + 2tP_{n-2}$, whose value at $t = 1$ is $2\big(a(n-1) + a(n-2)\big) > 0$, because at most one of $n-1$, $n-2$ equals 1. □

Remark 3 (reversed form; not formalized)**.** Let $Q_n(x) = x^n P_n(1/x) = \sum_{i=0}^{n} T(n, n-i)x^i = \sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}\sum_i E_2(j,i)x^i$. Taking $m = n+1$ in (3), substituting $t = 1/x$ and multiplying by x^{n+1} gives $Q_{n+1} = 2nx\,(Q_n + Q_{n-1}) + x(1-x)\,Q_n'$ for $n \geq 1$, with $Q_1 = 0$ and $Q_2 = 2x$. At $q = 1$ the recurrence of [7, Theorem 17] has the same initial values $D_1 = 0$, $D_2 = 2x$ and differs from this one only in the coefficient of the derivative term, $2x(1-x)$ instead of $x(1-x)$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A395454.lean`, reviewed for fidelity, and frozen (SHA-256 `99c82099...6646d`) before the proof was written. The definitions and the statement read as follows, inside `namespace Math0.Statements.A395454` with `open Polynomial` (verbatim; header, docstrings, theorems and the variant `statement_literal` omitted):

```
def E2 : ℕ → ℕ → ℤ
  | 0, 0 => 1
  | 0, _ + 1 => 0
  | n + 1, 0 => E2 n 0
  | n + 1, k + 1 =>
      (2 * ((n : ℤ) + 1) - 1 - ((k : ℤ) + 1)) * E2 n k + ((k : ℤ) + 1 + 1) * E2 n (k + 1)
def T (n k : ℕ) : ℤ :=
  ∑ j ∈ Finset.range (n + 1), (-1) ^ (n - j) * (n.choose j : ℤ) * E2 j (n - k)
def a (i : ℕ) : ℤ :=
  ((List.range (i + 1)).flatMap fun n => (List.range (n + 1)).map (T n)).getD i 0
noncomputable def P (n : ℕ) : ℤ[X] :=
  ∑ k ∈ Finset.range (n + 1), C (T n k) * X ^ k
noncomputable def P1 (n : ℕ) : ℤ[X] := P (n - 1)
def statement : Prop :=
  ∀ n : ℕ, 3 ≤ n →
    P1 n = ((n : ℤ[X]) - 2 * X + (n : ℤ[X]) * X - 2) * P1 (n - 1)
      + (X - X ^ 2) * derivative (P1 (n - 1))
      + X * (2 * (n : ℤ[X]) - 4) * P1 (n - 2)
```

Encoding. `E2` is (1), stated for row $n+1$ and computed in $\mathbb{Z}$. `T` is (2), Maple's $\binom{-j-1}{-n-1}$ being replaced by $(-1)^{n-j}\binom{n}{j}$; it is used only for $k \leq n$, where $n-k$ is exact. `a` is the triangle read by rows (the OEIS sequence, offset 0). `P` n is $P_n \in \mathbb{Z}[X]$ ($X = t$), `derivative` is the formal derivative, and `P1` $n = P_{n-1} = \widetilde{P}(n)$. The coefficients are cast to $\mathbb{Z}[X]$, and for $n \geq 3$ the natural-number subtractions in the indices are exact. `statement_literal` is the same formula with `P` for `P1`, for $n \geq 2$.

Checks against the data. The statement file proves `E2_data`, stating that the values `E2 n k` for $0 \leq k \leq n \leq 8$ are rows $0, \ldots, 8$ of A201637 (45 terms, written out), and `a_data`, stating that `(List.range 55).map (fun i => a (i + 0))` is the literal list of all 55 terms in the DATA section of A395454 (rows $0, \ldots, 9$). Both are proved by `decide +kernel`: the kernel evaluates the closed left-hand side by unfolding the definitions and compares it with the literal list. This anchors `E2`, `T` and `a` to the OEIS data independently of the conjecture. Finally, `not_statement_literal`: `¬ statement_literal` compares coefficients of t at $n = 2$, using `P_one`: `P 1 = 0` and `P_two`: `P 2 = C 2 * X`.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A395454.lean`; the Lean names of the corresponding declarations are given in brackets. All polynomials lie in $\mathbb{Z}[t]$, the prime denotes the formal derivative d/dt, and we write $u = -t$.

Lemma 4 (`E2_eq_zero`, `E2_succ_zero`, `E2_succ_succ`)**.** *For all $j, i \geq 0$: $E_2(j,i) = 0$ if $i > j$; $E_2(j+1, 0) = E_2(j, 0)$; and $E_2(j+1, i+1) = (2j - i)\,E_2(j,i) + (i+2)\,E_2(j, i+1)$.*

Proof. The last two are the defining clauses of `E2`, with simplified coefficients. The first follows by induction on j: $E_2(0,i)=0$ for $i\ge 1$, and if $i+1>j+1$ then $i>j$ and $i+1>j$, so both terms of $E_2(j+1,i+1)$ vanish. □

Lemma 5 (`XX_deriv_pow`, `XX_deriv_negpow`)**.** *For every $k\ge 0$, $(t-t^2)(t^k)' = k\,(t^k - t^{k+1})$ and $(t-t^2)(u^k)' = k(1-t)\,u^k$.*

Proof. Both sides vanish for $k=0$; for $k\ge 1$, $(t-t^2)kt^{k-1} = k(t^k-t^{k+1})$ and $-(t-t^2)ku^{k-1} = k(1-t)u^k$. □

For $j\ge 0$ define the *reversed second-order Eulerian polynomial* [`R`]

$$R_j(t) = \sum_{i=0}^{j} E_2(j,i)\,t^{j-i},$$

so $R_0=1$, $R_1=t$, $R_2=t^2+2t$, $R_3=t^3+8t^2+6t$.

Lemma 6 (`R_rhs`, `R_succ`)**.** *For every $j\ge 0$, $R_{j+1} = \big(j+(j+1)t\big)R_j + (t-t^2)R_j'$.*

Proof. For $i\le j$, Lemma 5 with $k=j-i$ gives $\big(j+(j+1)t\big)t^{j-i} + (t-t^2)(t^{j-i})' = (2j-i)t^{j-i} + (i+1)t^{j-i+1}$. Hence the right-hand side equals

$$\sum_{i=0}^{j}(2j-i)E_2(j,i)\,t^{j-i} + \sum_{i=0}^{j}(i+1)E_2(j,i)\,t^{j-i+1}. \tag{4}$$

On the left, split off the term $i=0$ of $R_{j+1} = \sum_{i=0}^{j+1} E_2(j+1,i)t^{j+1-i}$ and write the others with index $i+1$, $0\le i\le j$. By Lemma 4,

$$R_{j+1} = \sum_{i=0}^{j}(2j-i)E_2(j,i)\,t^{j-i} + \Big(\sum_{i=0}^{j}(i+2)E_2(j,i+1)\,t^{j-i} + E_2(j,0)\,t^{j+1}\Big).$$

The first sum is the first sum in (4). In the bracket the term $i=j$ vanishes, since $E_2(j,j+1)=0$ (Lemma 4); shifting the index of the remaining terms by one, the bracket becomes $\sum_{i=1}^{j}(i+1)E_2(j,i)t^{j-i+1} + 1\cdot E_2(j,0)t^{j+1}$, which is the second sum in (4). □

Next define the binomial transform [`S`]

$$S_n = \sum_{j=0}^{n}\binom{n}{j}u^{\,n-j}R_j \qquad (n\ge 0).$$

Lemma 7 (`S_term`, `P_eq_S`)**.** *For every $n\ge 0$, $P_n = S_n$.*

Proof. Let $0\le j\le n$. Since $(n-j)+(j-i) = n-i$ for $i\le j$,

$$\binom{n}{j}u^{n-j}R_j = \sum_{i=0}^{j}(-1)^{n-j}\binom{n}{j}E_2(j,i)\,t^{n-i} = \sum_{i=0}^{n}(-1)^{n-j}\binom{n}{j}E_2(j,i)\,t^{n-i},$$

where the sum may be extended to $i\le n$ because $E_2(j,i)=0$ for $i>j$ (Lemma 4). Summing over j and exchanging the two finite sums,

$$S_n = \sum_{i=0}^{n}\Big(\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}E_2(j,i)\Big)t^{n-i} = \sum_{k=0}^{n}\Big(\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}E_2(j,n-k)\Big)t^{k} = \sum_{k=0}^{n}T(n,k)t^k,$$

by the reflection $i=n-k$ of the range $\{0,\dots,n\}$ (`Finset.sum_range_reflect`) and (2). □

The recurrence for S_n is proved by writing every quantity as a sum over $0 \le j \le n+1$ of multiples of $\binom{n+1}{j}u^{n+1-j}R_j$ and $\binom{n+1}{j}u^{n+1-j}R_{j+1}$.

Lemma 8 (`S_succ`, `S_deriv`, `S_mul`)**.** *For every $n \ge 0$:*

$$\text{(a)}\quad S_{n+1} = \sum_{j=0}^{n} \binom{n}{j}\big(u^{n-j}R_{j+1} + u^{n-j+1}R_j\big);$$

$$\text{(b)}\quad (t-t^2)S_n' = \sum_{j=0}^{n} \binom{n}{j}\Big((n-j)(1-t)\,u^{n-j}R_j + u^{n-j}\big(R_{j+1} - (j+(j+1)t)R_j\big)\Big);$$

$$\text{(c)}\quad (n+1)\,u\,S_n = \sum_{j=0}^{n+1} \binom{n+1}{j}(n+1-j)\,u^{n+1-j}R_j.$$

Proof. (a) In $S_{n+1} = \sum_{j=0}^{n+1}\binom{n+1}{j}u^{n+1-j}R_j$ split off $j=0$ and write the other terms with index $j+1$, $0 \le j \le n$. By Pascal's rule $\binom{n+1}{j+1} = \binom{n}{j} + \binom{n}{j+1}$,

$$S_{n+1} = \sum_{j=0}^{n}\binom{n}{j}u^{n-j}R_{j+1} + \Big(\sum_{j=0}^{n}\binom{n}{j+1}u^{n-j}R_{j+1} + u^{n+1}R_0\Big).$$

In the bracket the term $j=n$ vanishes ($\binom{n}{n+1}=0$); shifting the index of the others, the bracket is $\sum_{j=0}^{n}\binom{n}{j}u^{n-j+1}R_j$.

(b) Differentiate termwise with the product rule, $(u^{n-j}R_j)' = (u^{n-j})'R_j + u^{n-j}R_j'$, and multiply by $t-t^2$. Lemma 5 with $k=n-j$ handles the first term, and Lemma 6, in the form $(t-t^2)R_j' = R_{j+1} - (j+(j+1)t)R_j$, the second.

(c) The term $j=n+1$ on the right is 0. For $j \le n$, $(n+1)\binom{n}{j} = (n+1-j)\binom{n+1}{j}$ (`Nat.choose_mul_succ_eq`) and $u\cdot u^{n-j} = u^{n+1-j}$, so the remaining terms agree with those of $(n+1)uS_n$. □

Proposition 9 (`S_rec`)**.** *For every $n \ge 0$,*

$$S_{n+2} = (n+1)(1+t)\,S_{n+1} + (t-t^2)\,S_{n+1}' + 2(n+1)t\,S_n.$$

Proof. Write $2(n+1)tS_n = -2\,(n+1)uS_n$. Apply Lemma 8(a) (with $n+1$ in place of n) to the left-hand side, the definition of S_{n+1} to the first term on the right, Lemma 8(b) (with $n+1$) to the second and Lemma 8(c) to the third. Every quantity is now a sum over $0 \le j \le n+1$, and it suffices to compare the summands. Fix j, put $K = n+1-j$, and divide by $\binom{n+1}{j}$. The summand on the left is $u^K R_{j+1} + u^{K+1}R_j = u^K R_{j+1} - t\,u^K R_j$, and on the right it is

$$u^K\Big(R_{j+1} + \big[(n+1)(1+t) + K(1-t) - j - (j+1)t - 2K\big]R_j\Big).$$

The terms with R_{j+1} agree, and since $n+1 = j+K$ the bracket equals $(j+K)(1+t) + K - Kt - j - jt - t - 2K = -t$. (In Lean: `Nat.cast_sub` for K, then `ring`.) □

Proof of Theorem 1. Let $n \ge 3$ and write $n = m+3$ with $m \ge 0$. Then $\widetilde{P}(n) = P_{m+2}$, $\widetilde{P}(n-1) = P_{m+1}$, $\widetilde{P}(n-2) = P_m$, and by Lemma 7 these equal S_{m+2}, S_{m+1}, S_m. The coefficients of the conjecture are $n-2t+nt-2 = (m+1)(1+t)$ and $t(2n-4) = 2(m+1)t$, so the claimed identity is Proposition 9 with m in place of n. Replacing $m+2$ by m gives (3). □

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A395454.lean` in the math0 repository, `https://github.com/Foundation-0/math0`; it imports the frozen statement file (Section 2) and `lean/Math0/Audit.lean`. The main declaration `Math0.Frontier.A395454.result` has type `Math0.Statements.A395454.statement`; the private auxiliary declarations are named in brackets in Section 3. As there, `result` writes $n = m + 3$, rewrites the three `P1` terms with `P_eq_S`, applies `S_rec m` and ends with `push_cast; ring`. Neither file uses `sorry`, `native_decide`, custom axioms or `set_option`; the statement file uses `decide +kernel` only for its data checks and for the values of `T` used in `P_one` and `P_two`.

Audit. The file ends with `#math0_settles Math0.Frontier.A395454.result Math0.Statements.A395454.statement`. This command (`lean/Math0/Audit.lean`) fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) both declarations depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (so no `sorryAx`, `native_decide` or custom axioms); and (3) the type of `result` is syntactically `statement` or $\neg$ `statement`, up to metadata, with no unfolding. It reports

```
math0 audit: 'Math0.Statements.A395454.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A395454.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A395454.result' proves 'Math0.Statements.A395454.statement'
```

Toolchain and reproduction. We use Lean 4 [5] `v4.33.1` and Mathlib [8] `v4.33.1`, pinned in `lean/`; formal-conjectures [2] is pinned at commit `c77fdd4` but not used here. Running `scripts/verify.sh` from the repository root fetches the Mathlib cache, rebuilds all results with `lake build` (re-running every audit), checks that the pinned dependencies are unmodified, and counts the passed statement checks. The single file is checked by `lake env lean Math0/Frontier/OEIS/A395454.lean` in `lean/`.

References

[1] G.-H. Duh, Y.-C. R. Lin, S.-M. Ma, and Y.-N. Yeh, *Stirling permutations, marked permutations and Stirling derangements*, arXiv:1612.07391 [math.CO], 2016.

[2] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171 [cs.AI], 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[3] S. Fried, *Further proofs of conjectures from the OEIS*, arXiv:2607.24832 [math.GM], 2026.

[4] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics: A Foundation for Computer Science*, 2nd ed., Addison-Wesley, Reading, MA, 1994.

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[6] S.-M. Ma, S. Kitaev, J. Yeh, and Y.-N. Yeh, *Eulerian-type polynomials over matchings and matching permutations*, arXiv:2601.20665 [math.CO], 2026.

[7] S.-M. Ma and Y.-N. Yeh, *Eulerian polynomials, Stirling permutations of the second kind and perfect matchings*, Electron. J. Combin. **24**(4) (2017), #P4.27, `https://doi.org/10.37236/7288`; arXiv:1607.01311.

[8] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[9] The OEIS Foundation Inc., entry A053871 (C. Moore and C. G. Bower, 2000) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A053871`.

[10] The OEIS Foundation Inc., *Triangle of second-order Eulerian numbers*, entry A201637 (P. Luschny, 2012) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A201637`.

[11] The OEIS Foundation Inc., entry A395454 (P. Luschny, 2026; conjecture by P. W. Alexandersson, June 27, 2026) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A395454`; revision #30, October 14, 2026.

Produced by Claude (an AI model developed by Anthropic) in the math0 project